Offset-Free Model Predictive Control for an Induction Heater: Comparison with Hand-Tuned and Optimised PID

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Abstract

Model predictive control (MPC) plans the next actuator move by rolling a plant model forward over a finite horizon and solving a constrained optimisation at every sample. In this report we build a simple offset-free MPC for an induction heater, run it on a CasADi heat-balance simulator, and compare it with a hand-tuned PID and a SciPy-optimised PID on the same tests. We score tracking with integrated absolute error (IAE) and also count power-limit violations. After a short retune (let the horizon see the future setpoint, N=40, terminal weight Q_t=50), MPC has the best IAE on setpoint tracking (1044 vs 1466 vs 2477), under tight power limits (8727 vs 9033 vs 9153), and under model mismatch (1214 vs 1667 vs 2541), with no limit violations when constraints are active. Optimised PID is slightly better on two disturbance tests when it is tuned on those same tests. Overall, MPC is the better choice for this heater: it respects coil power and slew limits by design, recovers faster when the model is wrong, and matches or beats PID on the main tracking cases once it can see the coming setpoint.

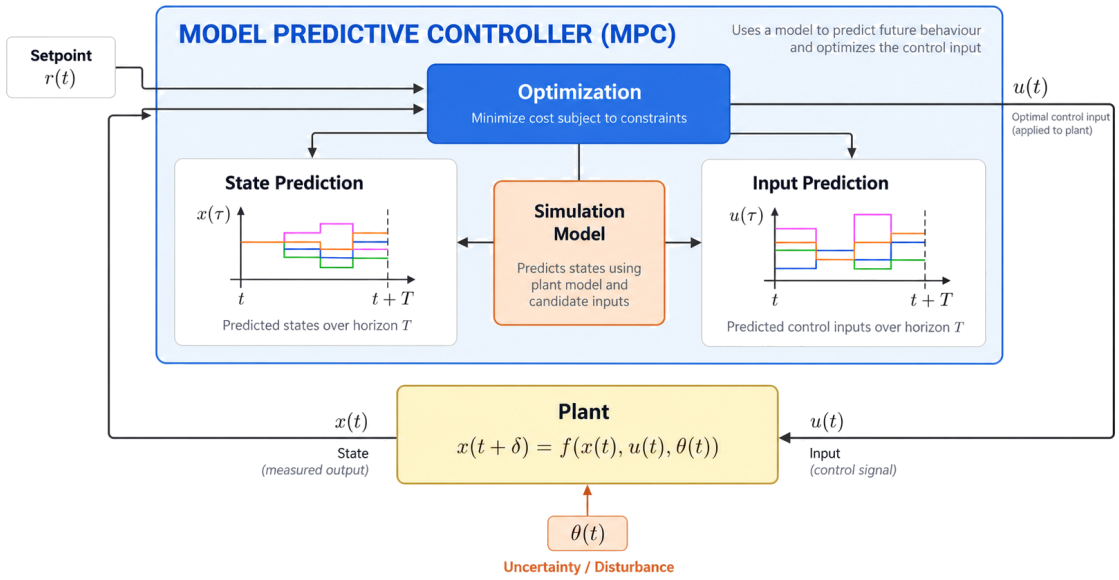


Figure 1: MPC loop: predict states and inputs over a horizon, apply the first move to the plant, and close the loop with measured feedback and disturbance $\theta(t)$.

1 Introduction

Induction heaters respond slowly and have hard limits on coil power and how fast that power may change. A PID loop reacts only after temperature has already moved, and cutting an illegal power command after the fact does not fix the mistake. MPC uses a model to predict what temperature will do under different power choices, re-solves every sample (receding horizon), and can enforce

power and rate limits inside the optimiser [2, 3]. Figure 1 shows the idea: predict over a short horizon, apply only the first chosen move, then measure the plant and repeat.

We implement that idea with a fitted linear model plus a disturbance estimate for offset-free tracking [1], a SciPy solver, a CasADi heater simulator as the truth plant, batch metrics, and a three-screen Streamlit lab. The physical numbers in this report are illustrative coursework values, not field data.

The practical question is: after we retune the MPC settings (Section 3.1), and after we let an optimised PID see each test during tuning, which controller should run this limited heater?

2 Background

2.1 Heater plant

The plant is a simple heat balance on workpiece temperature T:

$$mc_p \frac{dT}{dt} = \eta P_{\text{coil}} - U\big(T - T_{\text{amb}}\big), \tag{1}$$

integrated with explicit Euler in CasADi. Demo parameters are m=800 kg, c_p=480 J kg^{-1} K^{-1}, η=0.82, U=45 W K^{-1}, T_{amb}=25 °C.

2.2 Identified prediction model

MPC predicts with a one-step SISO ARX model

$$y_{k+1} = a y_k + b u_k + \hat{d}_k, \tag{2}$$

with a=0.978, b=0.042, Δt=1 s from a 120 s least-squares step fit on the simulator (RMS residual 0.35 °C). To remove steady-state offset we use the integrating disturbance of Pannocchia and Rawlings [1]. After the first sample,

$$\hat{d}_k \leftarrow \hat{d}_{k-1} + g\big(y_k^{\text{meas}} - \hat{y}_{k|k-1}\big), \tag{3}$$

with observer gain g=1 in the retuned controller.

2.3 MPC cost, constraints, and receding horizon

Over horizon N the controller chooses one free move u (control horizon N_u=1 after the sweep in Section 3.1) and minimises

$$J(u) = \sum_{i=1}^{N} \big(y_i - r_i\big)^2 + Q_t\big(y_N - r_N\big)^2, \tag{4}$$

where r_i is the future setpoint seen over the horizon (reference preview). The chosen u must stay inside the coil power window and the one-step slew limit:

$$\max(u_{\min},\, u_0 - \Delta u_{\max}) \le u \le \min(u_{\max},\, u_0 + \Delta u_{\max}). \tag{5}$$

Only the first move is applied; then the horizon shifts forward. We solve the problem with SciPy `L-BFGS-B`.

2.4 PID baseline

The discrete PID is

$$u_k = u_{k-1} + K_p e_k + K_i \sum e + K_d \frac{e_k - e_{k-1}}{\Delta t}, \tag{6}$$

with back-calculation anti-windup when the actuator saturates [4]. Hand-tuned gains are the scenario defaults in the repository. Optimised PID uses SciPy `L-BFGS-B` to minimise IAE + 250 × (violation steps) on the same plant, noise seed, and limits, with bounds $K_p \in [0.05, 12]$, $K_i \in [0, 0.35]$, $K_d \in [0, 8]$.

3 System design and implementation

Figure 2 shows the closed loop. MPC, hand-tuned PID, and optimised PID all drive the same CasADi simulator and are scored with the same metrics: IAE, ISE, overshoot, ±3% settling time, constraint-violation steps, and energy $\int P\,dt$.

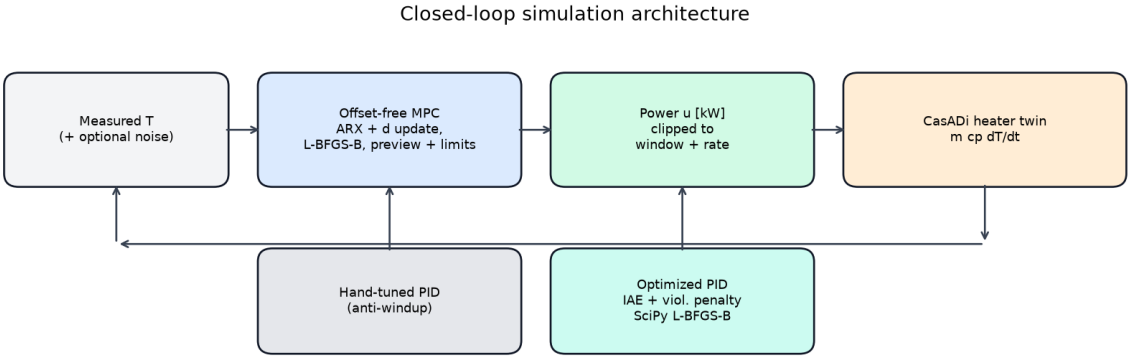
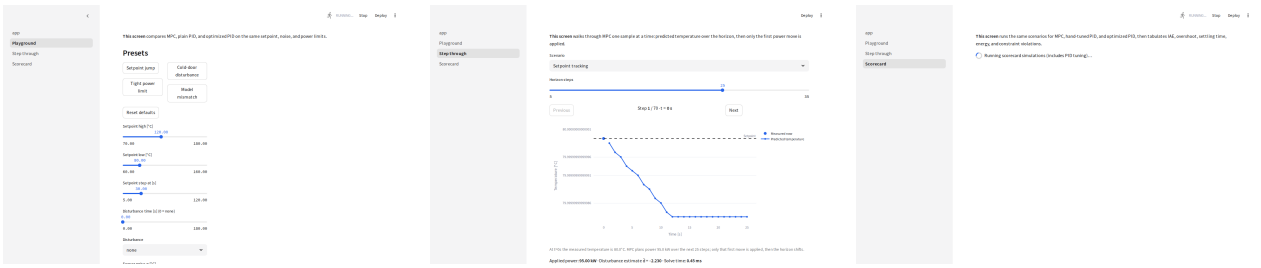


Figure 2: Simulation architecture: offset-free MPC and both PID variants act on one heater twin.

The repository also includes a Streamlit lab (Playground, Step-through, Scorecard) that uses the same presets as the batch metrics below. Figure 3 shows those three screens.



(a) Playground **(b)** Step-through **(c)** Scorecard

Figure 3: Interactive lab screens used to inspect MPC against both PID baselines.

3.1 Retuned MPC parameters

We ran a short grid on setpoint tracking, starting from legacy settings (N=25, N_u=1, no preview, Q_t=0) and trying longer horizons, multi-move N_u, terminal weights, move penalties, and observer gains. Allowing several free moves with a move penalty made IAE worse on this heater. What helped most was **reference preview**, a longer horizon, and a terminal weight. The retuned values used for all results below are:

Knob	Legacy	Retuned
Prediction horizon N	25	40 (45 in power-limit case)
Control horizon N_u	1	1
Terminal weight Q_t	0	50
Move weight R	0	0
Observer gain g	1	1
Reference preview	no	yes
Setpoint-tracking IAE	1824.3	1044.3

Table 1: MPC settings before and after the retune.

These settings live in the repository defaults under `src/mpc_core/` (`mpc.py`, `sim/config.py`).

4 Experimental setup

Sample time Δt=1 s. Default power window 0–250 kW with slew 25 kW s^{-1}. We use five scenarios:

1. **Setpoint tracking:** 80 → 120 °C at 30 s (duration 180 s).
2. **Disturbance rejection:** hold 115 °C; heat-loss U rises 45% at 70 s.
3. **Power limits:** cap 150 kW, slew 8 kW s^{-1}, target 165 °C. Hand-tuned PID has rate limiting off (as in the repository scenario) to show the cost of ignoring slew; MPC and optimised PID stay within limits.
4. **Model mismatch:** same setpoint test, but plant mass +8%, efficiency −6%, heat loss +18%. MPC still uses the nominal ARX model.
5. **Ambient step:** T_{amb} rises by 8 K at 70 s.

We also run a robustness check: keep optimised-PID gains from the normal setpoint plant, then switch to the mismatched plant, and compare with MPC and with a PID retuned on the mismatch itself.

5 Results

Blue bold cells mark where MPC is best among the three controllers. Other winners stay in plain type.

5.1 Summary table

Table 2: Closed-loop metrics after the MPC retune. Blue bold = MPC best.

Scenario	Metric	MPC	PID	Opt. PID
Setpoint	IAE	**1044.3**	2476.9	1466.4
	Settling [s]	**73**	179	99
	Violations	**0**	0	0
Disturbance	IAE	596.0	1089.9	585.8
	Settling [s]	**46**	103	60
	Violations	**0**	0	0
Power limits	IAE	**8727.4**	9153.3	9032.6
	Violations	**0**	6	0
	Energy [kWh]	8.297	7.986	8.073
Mismatch	IAE	**1214.2**	2541.2	1666.6
	Settling [s]	**83**	167	109
Ambient step	IAE	641.2	1092.4	578.3
	Settling [s]	**46**	102	60

Figure 4 summarises IAE; hatched bars are MPC wins and match Table 2. Figure 5 shows the four main time plots behind that table: MPC reaches the 120 °C step first, stays inside the tight power cap, and recovers well under mismatch, while optimised PID stays close on the heat-loss disturbance.

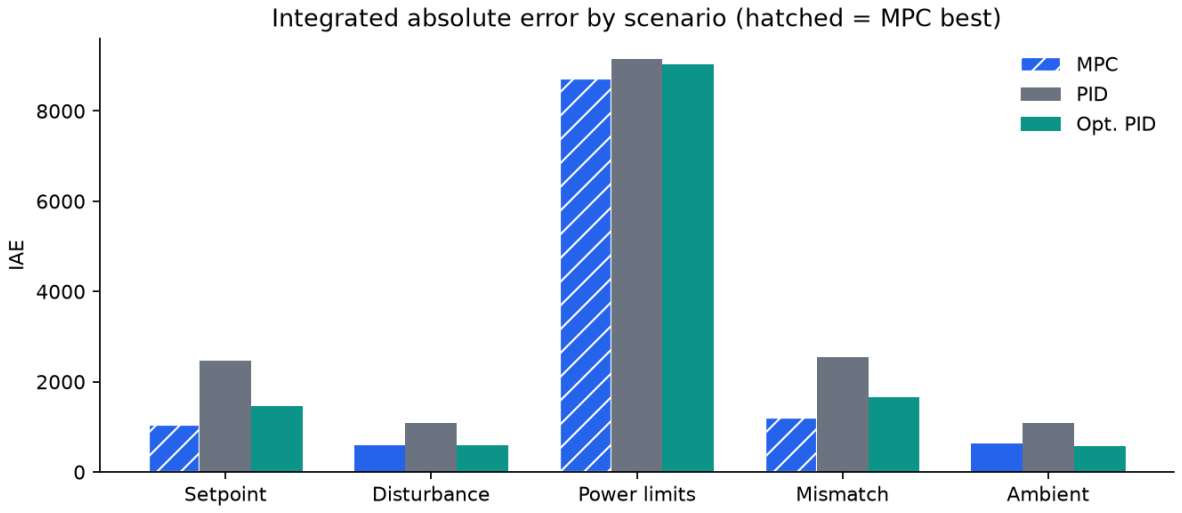
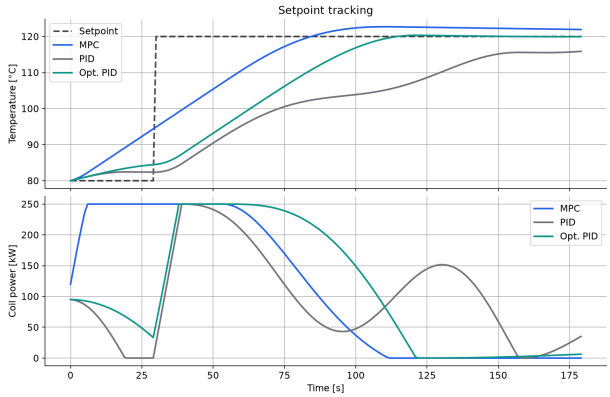
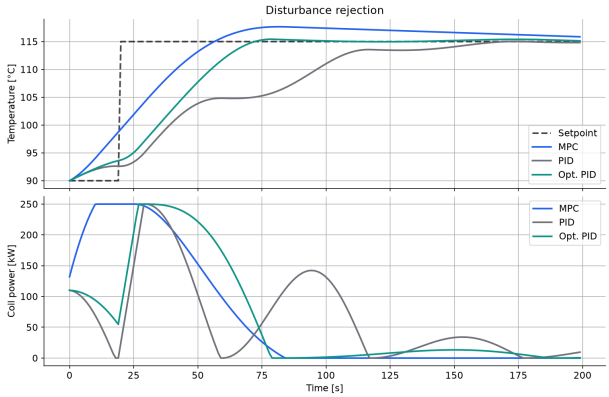


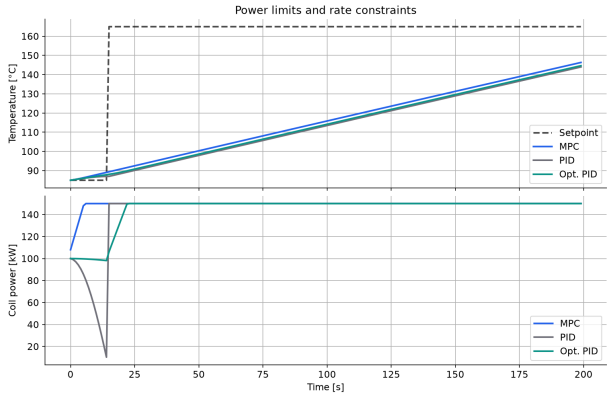
Figure 4: IAE by scenario. Hatched bars mark where MPC is lowest.



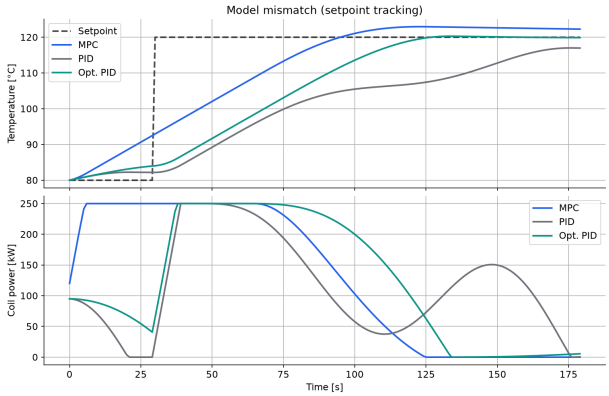
(a) Setpoint tracking



(b) Disturbance rejection



(c) Power limits



(d) Model mismatch

Figure 5: Temperature and coil power for MPC, PID, and optimised PID (aligned with Table 2).

5.2 Power limits and violations

On the hard 165 °C request with a 150 kW / 8 kW s^{-1} ceiling, hand-tuned PID makes **six** slew violations and only looks slightly cheaper in energy because those illegal moves are allowed. MPC

and optimised PID stay at zero violations; MPC also has the lowest IAE (8727). Figure 6 shows the gap. Ambient-step IAE and settling in Table 2 follow the same pattern as Figure 5(b): optimised PID wins IAE after a scenario retune, while MPC still settles faster because $\hat{d}$ absorbs the load change.

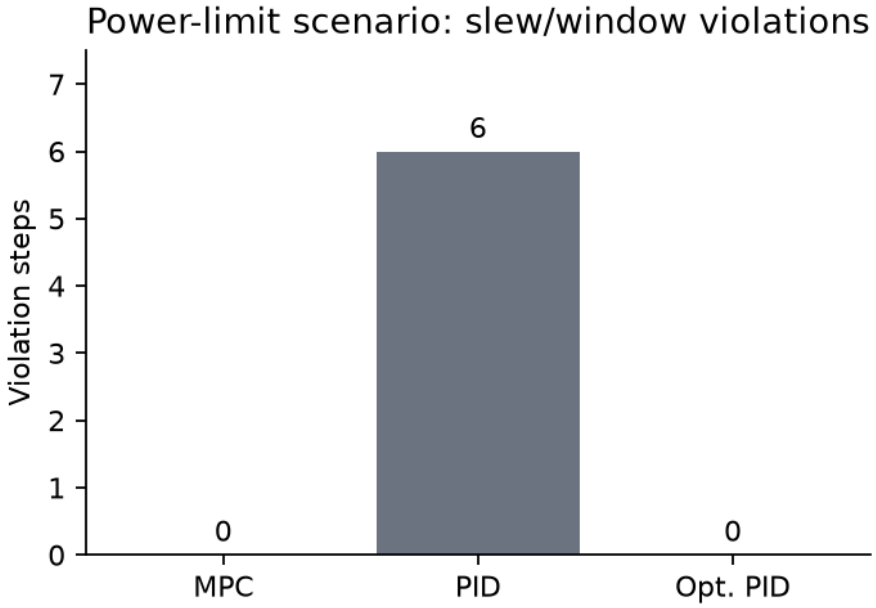


Figure 6: Constraint-violation counts on the power-limit scenario (PID has six slew breaches; MPC and optimised PID stay within limits).

5.3 Robustness

If we freeze optimised-PID gains from the normal setpoint test and then switch to the mismatched plant, IAE rises to 1702. MPC on the same mismatched plant stays at 1214. Even a PID retuned on the mismatch itself only reaches 1667. Figure 7 shows the comparison and matches the mismatch bars in Figure 4.

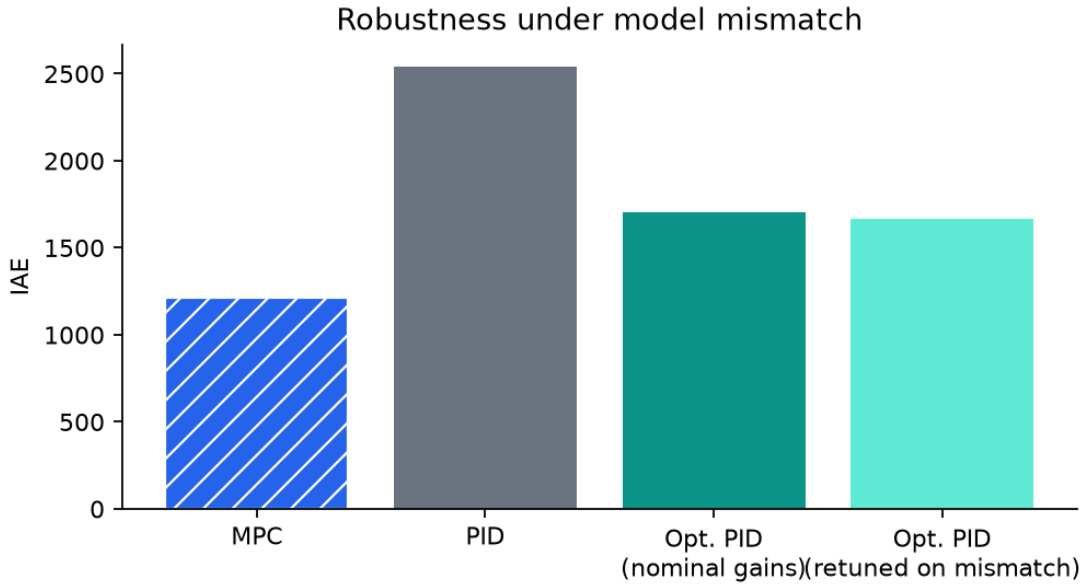


Figure 7: IAE under model mismatch. MPC stays lowest even against a PID retuned on the mismatched plant.

6 Discussion and limitations

For a coil that must stay inside a legal power window, the retuned MPC is the better controller. It has the lowest tracking IAE on setpoint, power-limit, and mismatch tests (Figures 4–5), the fastest settling on every row in Table 2, zero violations when limits bind (Figure 6), and a clear edge when the plant drifts from the model used to tune the PID (Figure 7).

Two IAE rows still favour scenario-tuned optimised PID: disturbance rejection (586 vs MPC 596) and the ambient step (578 vs 641). Those gains were fitted on the same disturbance the controller then faces, so this is a strong and fair baseline. MPC still settles faster in both cases (46 s vs 60 s) because $\hat{d}$ absorbs the load change without a retune. Hand-tuned PID is never best on IAE or settling here; its only "win" on energy in the power-limit case comes with six illegal slew steps.

This coursework controller still has limits: only one free move (N_u=1), a linear predictor against a nonlinear plant, power constraints only (no temperature constraint), and illustrative parameters. A production design would add move blocking, multi-input coupling, and a clearer model check before recommending a move. The Streamlit lab is for teaching, not a PLC programme.

7 Conclusion

For the induction heater studied here, offset-free MPC with reference preview is the controller to prefer. After the retune it leads on the main tracking and constraint metrics, keeps the coil inside limits by design, and stays ahead of both a frozen and a retuned PID when the plant is mismatched. Where a scenario-tuned PID still posts a slightly lower disturbance IAE, the gap is small, and that PID needs offline retuning that MPC does not. The retuned settings (N=40, Q_t=50, preview on) are the defaults used for the figures in this report.

References

[1] G. Pannocchia and J. B. Rawlings, "Disturbance models for offset-free model predictive control," *AIChE Journal*, vol. 49, no. 2, pp. 426–437, 2003.

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[5] P. Virtanen et al., "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

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